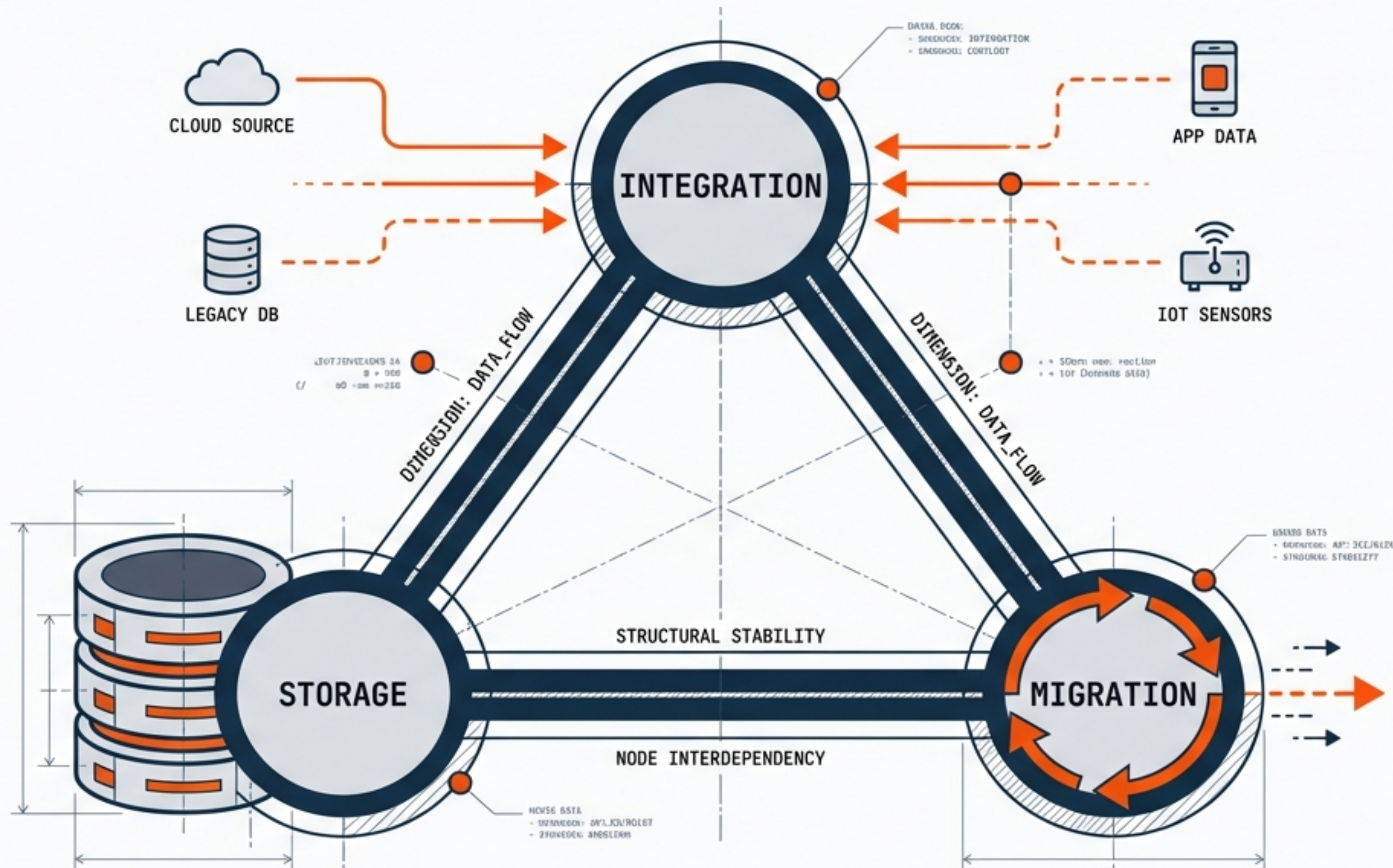


THE PILLARS OF MODERN DATA MANAGEMENT

A Strategic Overview of Integration, Storage, and Migration



DATA MANAGEMENT IS AN ENGINE FOR GROWTH



Managing data is no longer just about IT maintenance; it is about **building the infrastructure for business intelligence.**



SECURITY
PROTOCOL



As organizations face **'data chaos'**—**silos, security risks,** and **legacy debt**—the demand for professionals who can unify, secure, and modernize data assets is critical.

DATA SILO 01

SECURITY
ADJUSTMENTS

LEGACY SYSTEM DEBT



Source: IoT Analytics

DWG NO.
DATA-MGMT-GROWTH-001

SCALE
1:1

REV
A

ENGINEERING DATA VISUALIZATION

DATE:
OCT 26, 2024

THE DATA LIFECYCLE ARCHITECTURE



Source: IoT Analytics

DWG NO.
DATA-MGMT-GROWTH-002

SCALE
1:1

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ENGINEERING DATA VISUALIZATION

DATE:
OCT 26, 2024

01. INTEGRATE

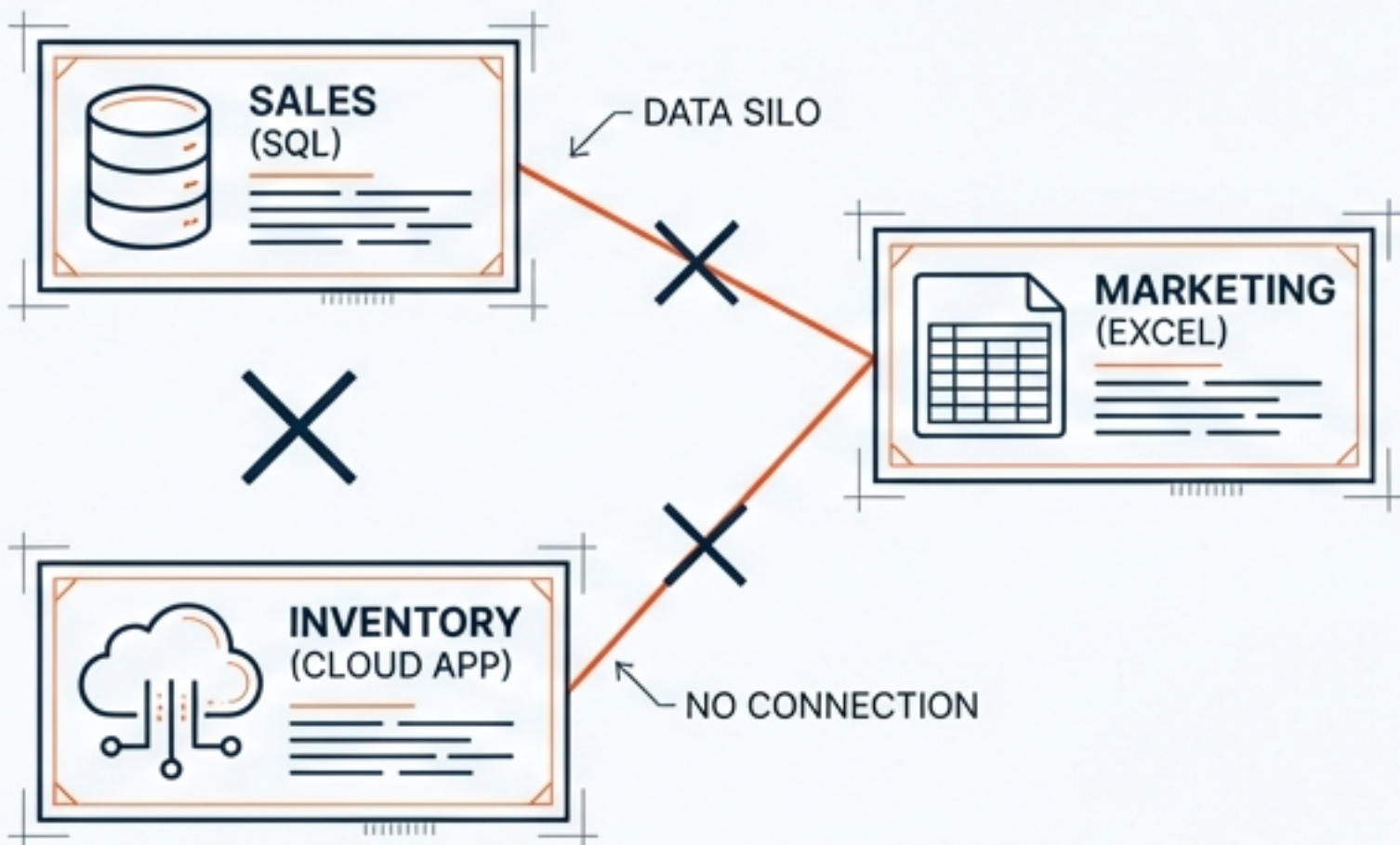
02. STORE

03. MIGRATE

AGGREGATING DISPARATE SOURCES

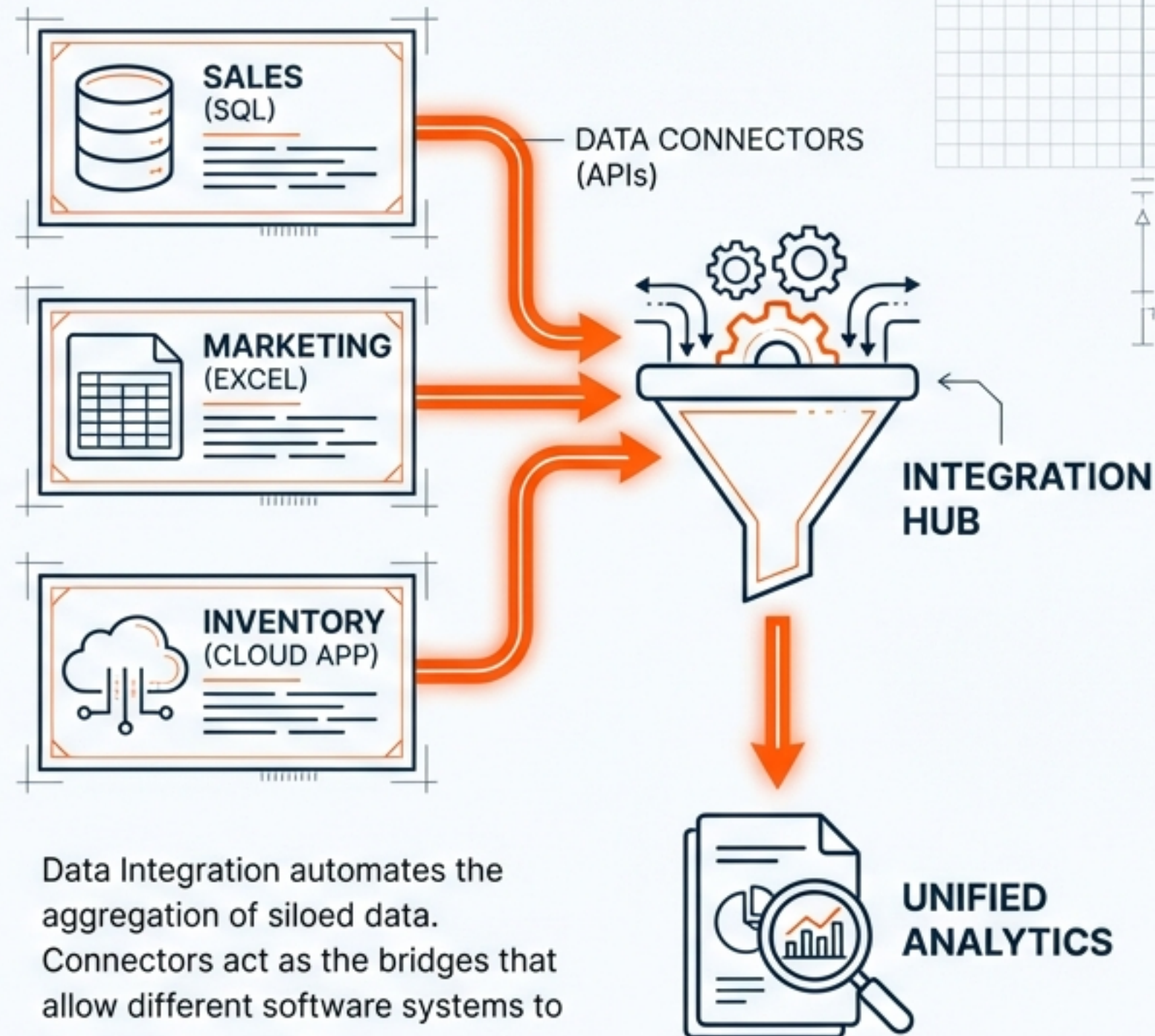
Breaking down silos to create a unified view

THE CHALLENGE



Source: IoT Analytics

THE SOLUTION

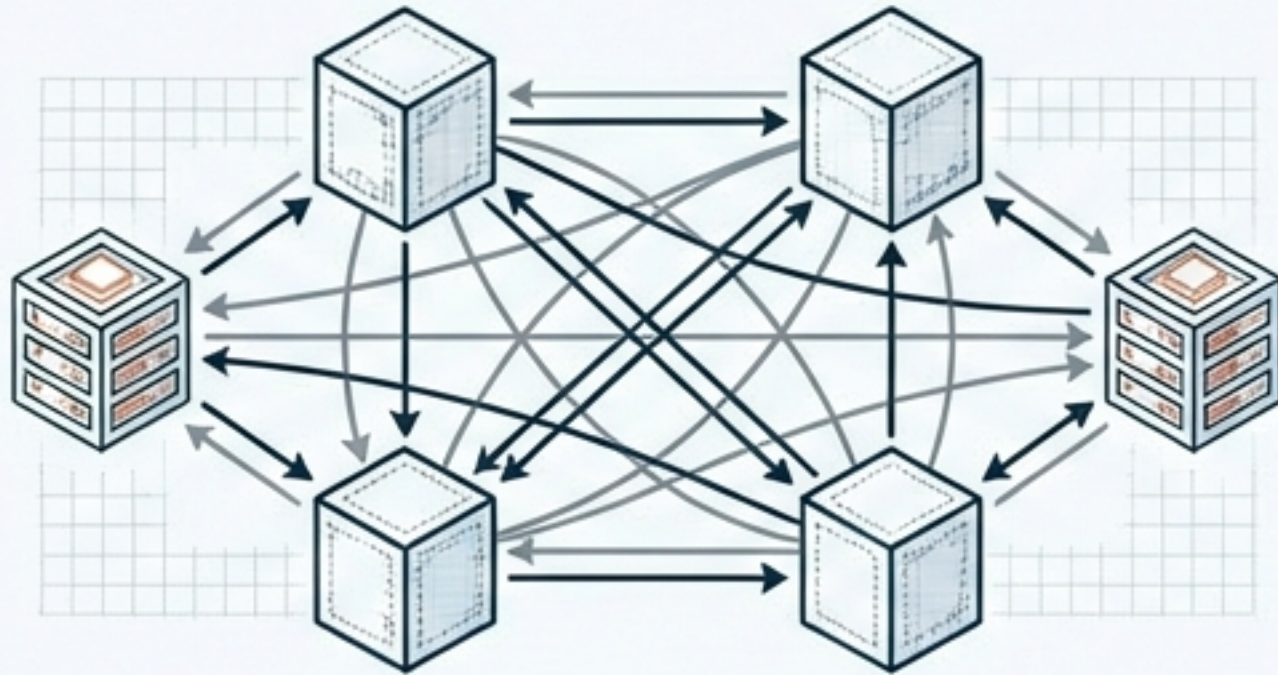


Data Integration automates the aggregation of siloed data. Connectors act as the bridges that allow different software systems to speak to one another.

DWG NO.	DATA-MGMT-GROWTH-003	SCALE	1:1	REV	A
	ENGINEERING DATA VISUALIZATION	DATE:	OCT 26, 2024		

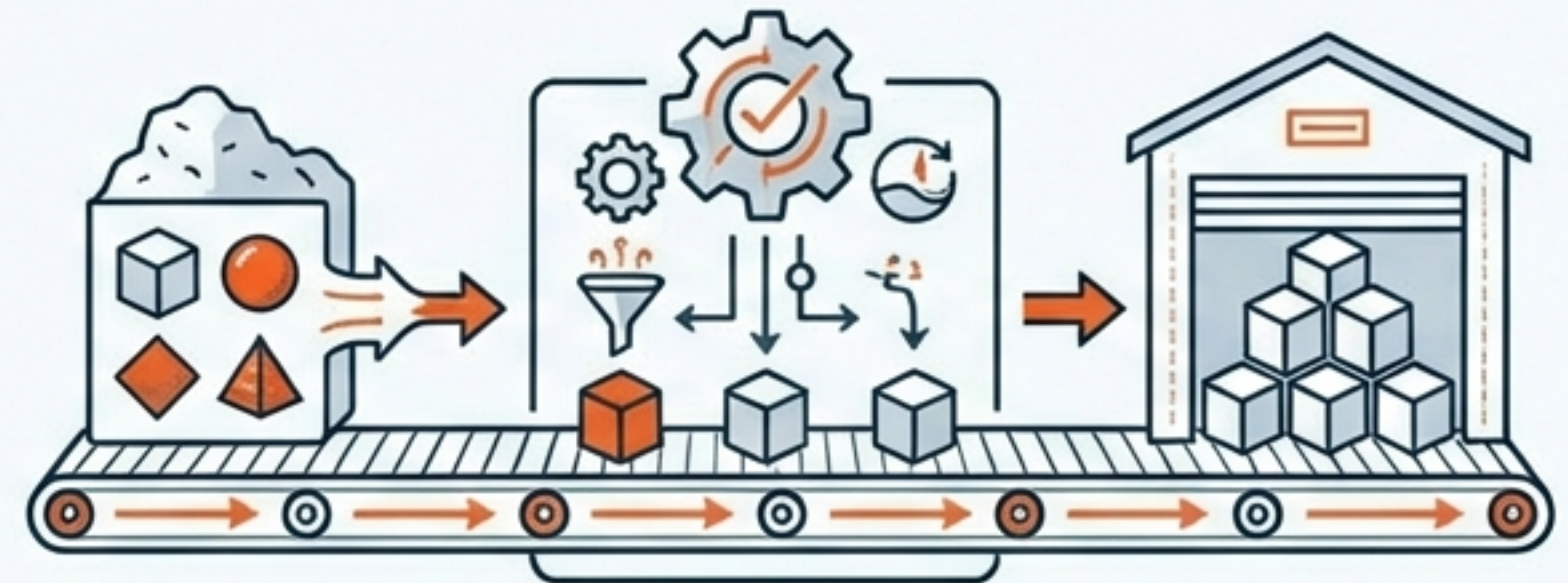
CHOOSING AN INTEGRATION PATTERN

POINT-TO-POINT (P2P)



Direct connections between systems. Simple to start, but hard to scale and maintain.

ETL (EXTRACT, TRANSFORM, LOAD)



EXTRACT

TRANSFORM/CLEAN

LOAD

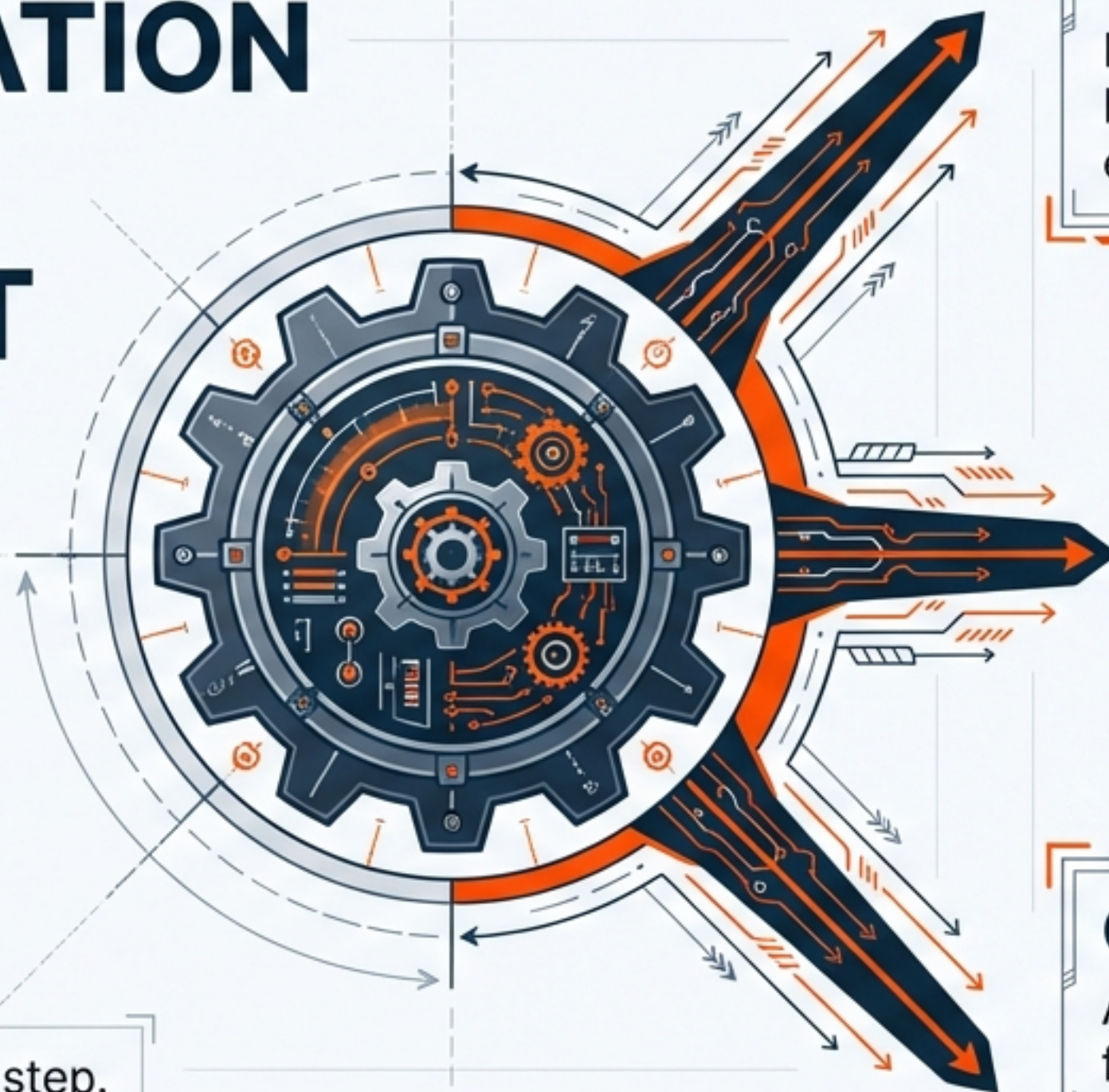
The industry standard for warehousing. Data is cleaned before storage, ensuring quality and consistency.

01. INTEGRATE

02. STORE

03. MIGRATE

DATA INTEGRATION OPERATIONS MANAGEMENT (DIOM)



SECURITY



Protecting data in transit
between connectors and
endpoints.

ARCHITECTURE



Designing scalable flows
rather than ad-hoc fixes.

ORCHESTRATION



Automating the schedule and
flow of data pipelines.



Connecting data is only the first step.
DIOM ensures the pipeline remains
secure, efficient, and operational.

D8A

DWG NO.
DATA-MGMT-GROWTH-005

SCALE
1:1

REV
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ENGINEERING DATA VISUALIZATION

DATE: OCT 28, 2024

ARCHITECTURES FOR DATA PERSISTENCE

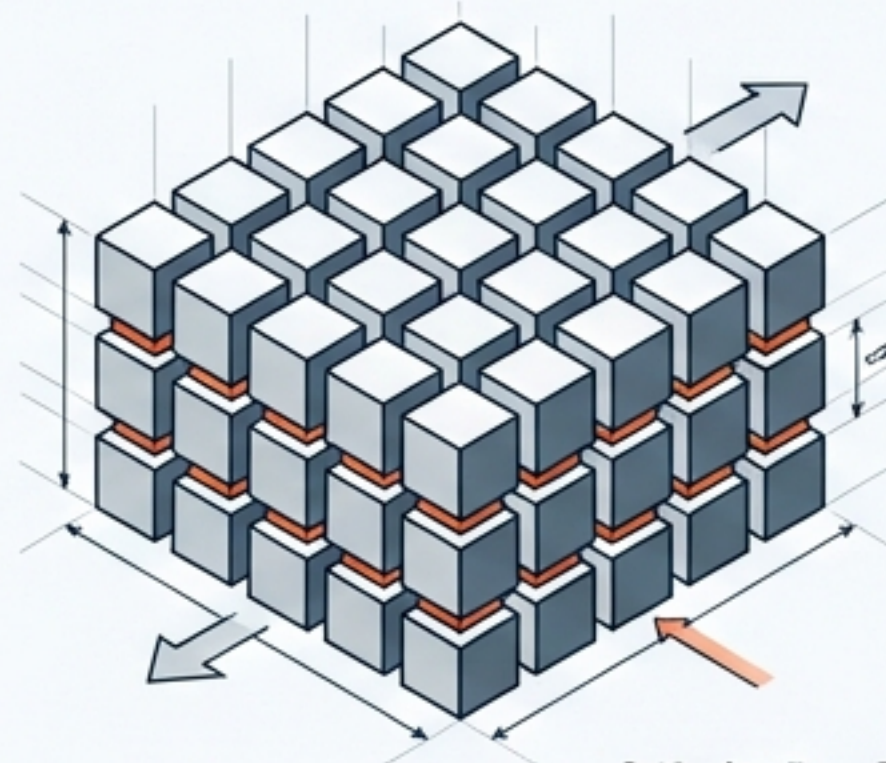
FILE STORAGE



JetBrains Rono, Inter

Hierarchical folders.
Best for general computing.

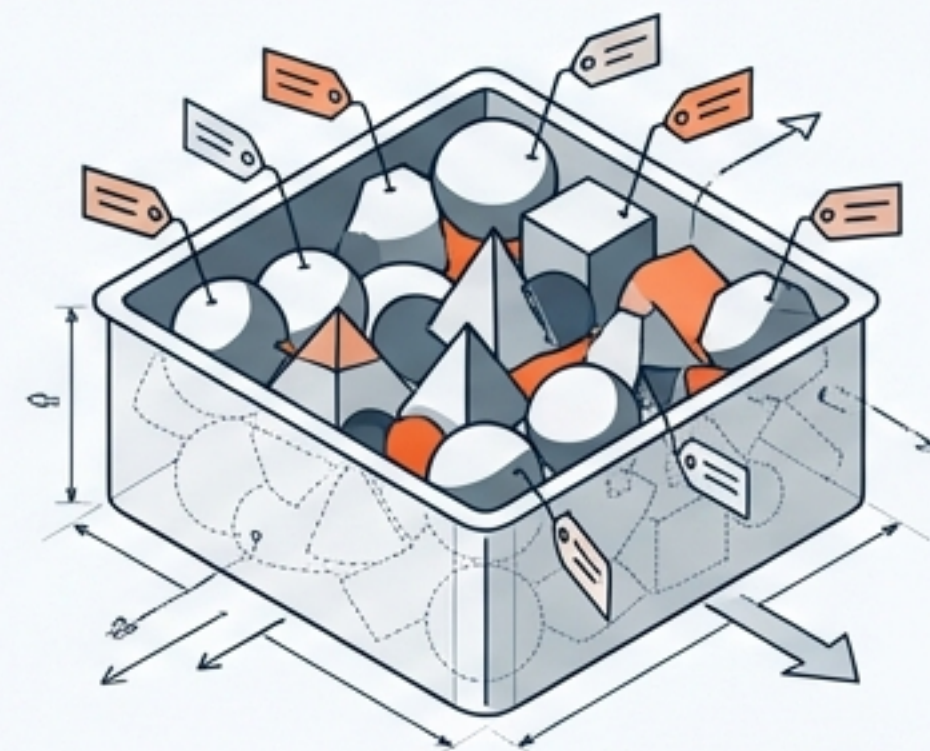
BLOCK STORAGE



JetBrains Rono, Inter

Fixed-size chunks.
High performance. Ideal for databases.

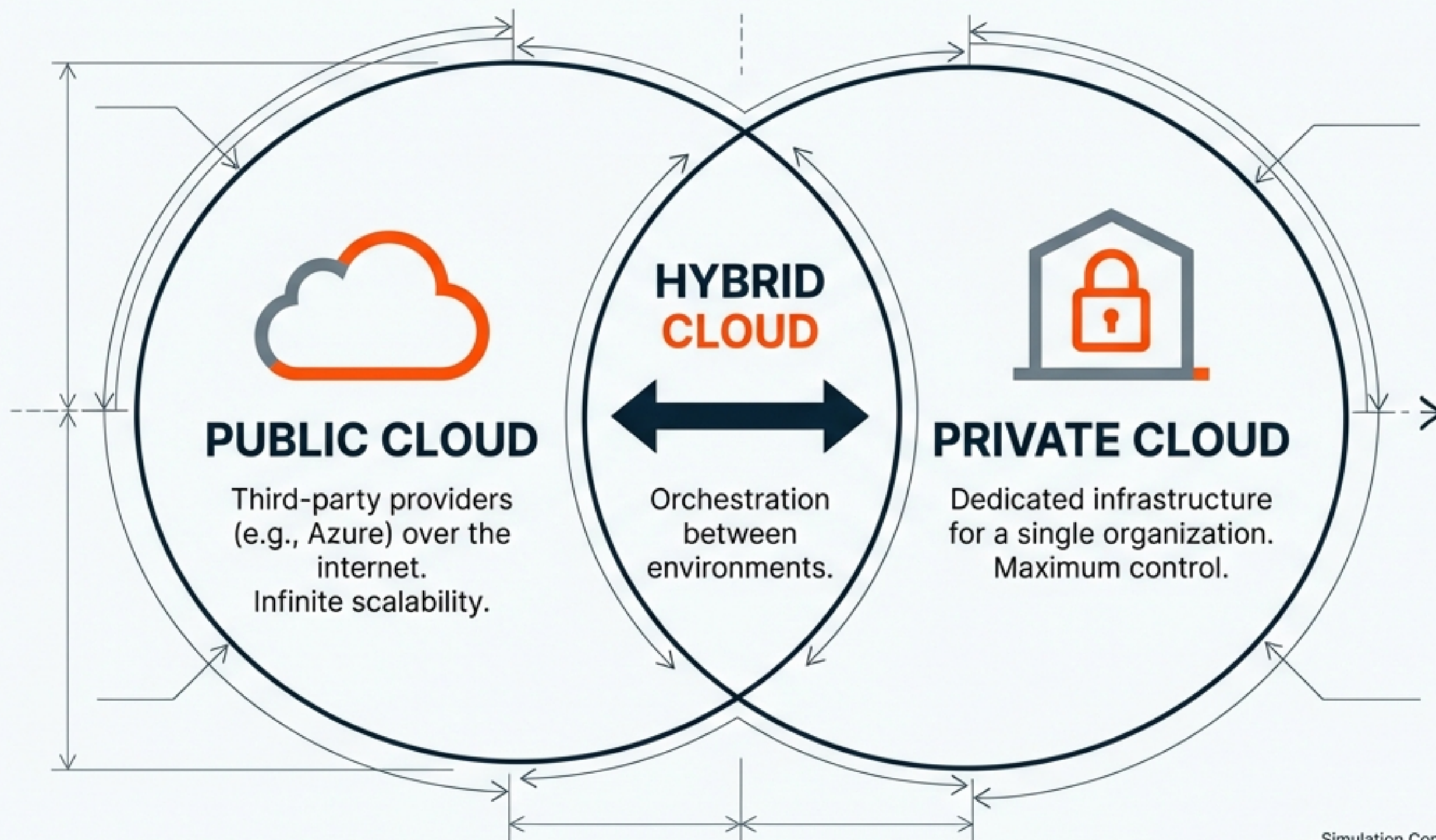
OBJECT STORAGE



JetBrains Rono, Inter

Scalable data with metadata.
Ideal for unstructured media.

THE UBIQUITY OF CLOUD STORAGE



Simulation Context
Simulation Context details



**MODERN
STORAGE**



Modern storage has moved beyond the physical data center, utilizing services like Azure to offer scalability on-premise hardware cannot match.

Technical Editorial

Modern Architectural Blueprint

DWG NO.
DATA-MGMT-GROWTH-007

SCALE:
1:1

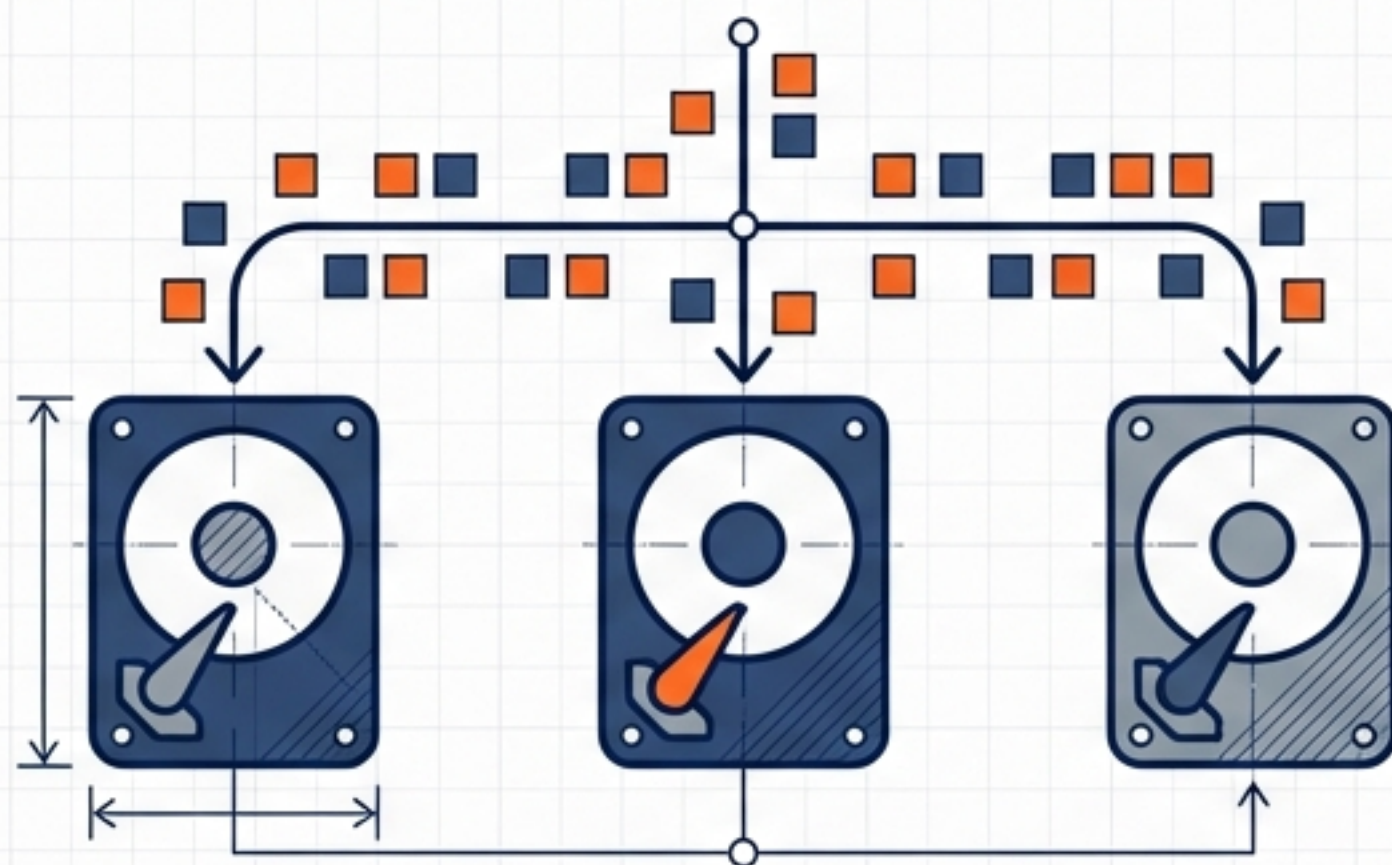
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ENGINEERING DATA
VISUALIZATION

DATE:
OCT 28, 2024

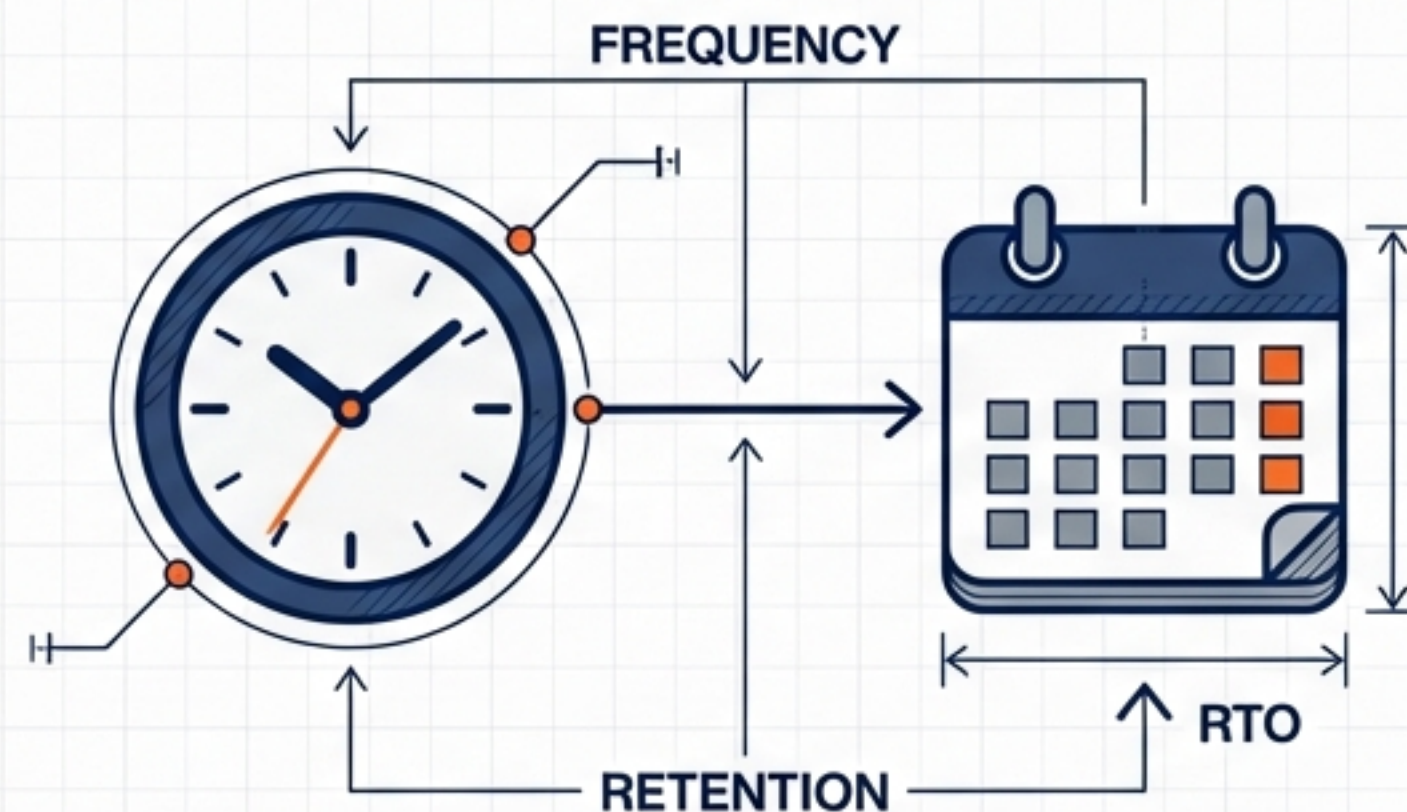
IF IT ISN'T BACKED UP, IT DOESN'T EXIST.

A RAID (Redundancy)



Hardware failure protection via redundancy.

B BACKUP STRATEGY



Defining Frequency, Retention, and Recovery Time Objectives (RTO).

Technical Editorial

DWG NO. DATA-MGMT-GROWTH-008

SCALE: 1:1

REV: A

Modern Architectural Blueprint

ENGINEERING DATA VISUALIZATION

DATE: OCT 28, 2024

MODERNIZING VIA DATA MIGRATION



LEGACY SYSTEMS

JetBrains Mono

Migration is the process of moving data to upgrade infrastructure. The goal is modernization with minimal disruption to business operations.

MODERN DATA LAKE

JetBrains Mono

Simulation Context
Simulation Context actais



**MODERN
STORAGE**



Modern data lake is enhances a stream structure, vast storage, advanced analytics, and seamless connectivity.

Technical Editorial

Modern Architectural Blueprint

DWG NO.
DATA-MGMT-GROWTH-009

SCALE:
1:1

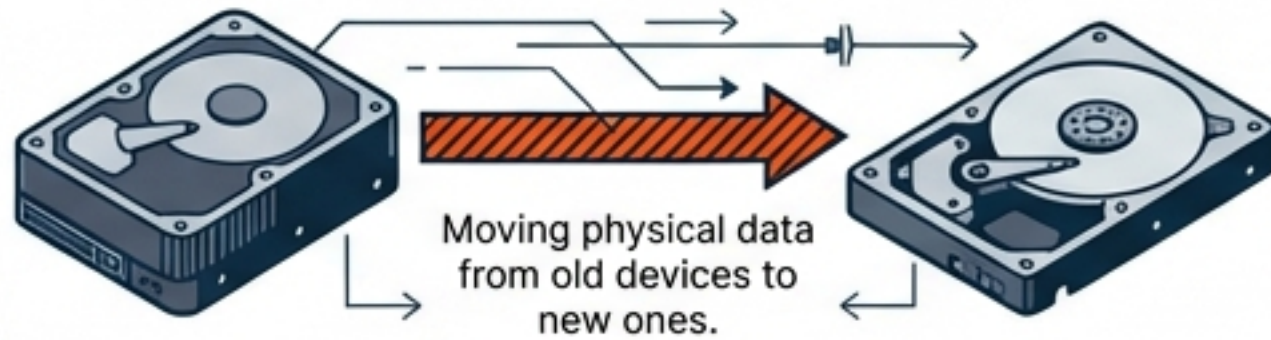
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ENGINEERING DATA
VISUALIZATION

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OCT 28, 2024

TYPES OF DATA MOVEMENT

1. STORAGE MIGRATION



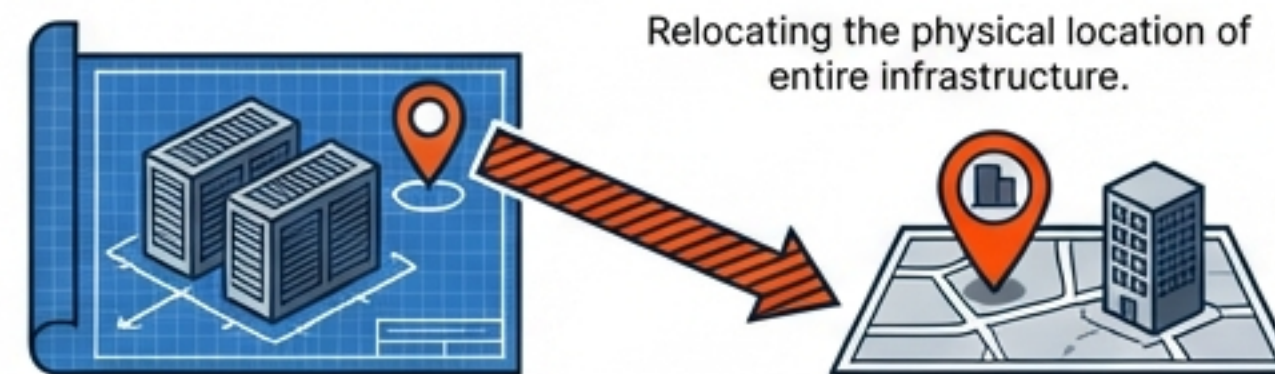
2. CLOUD MIGRATION



3. APPLICATION MIGRATION

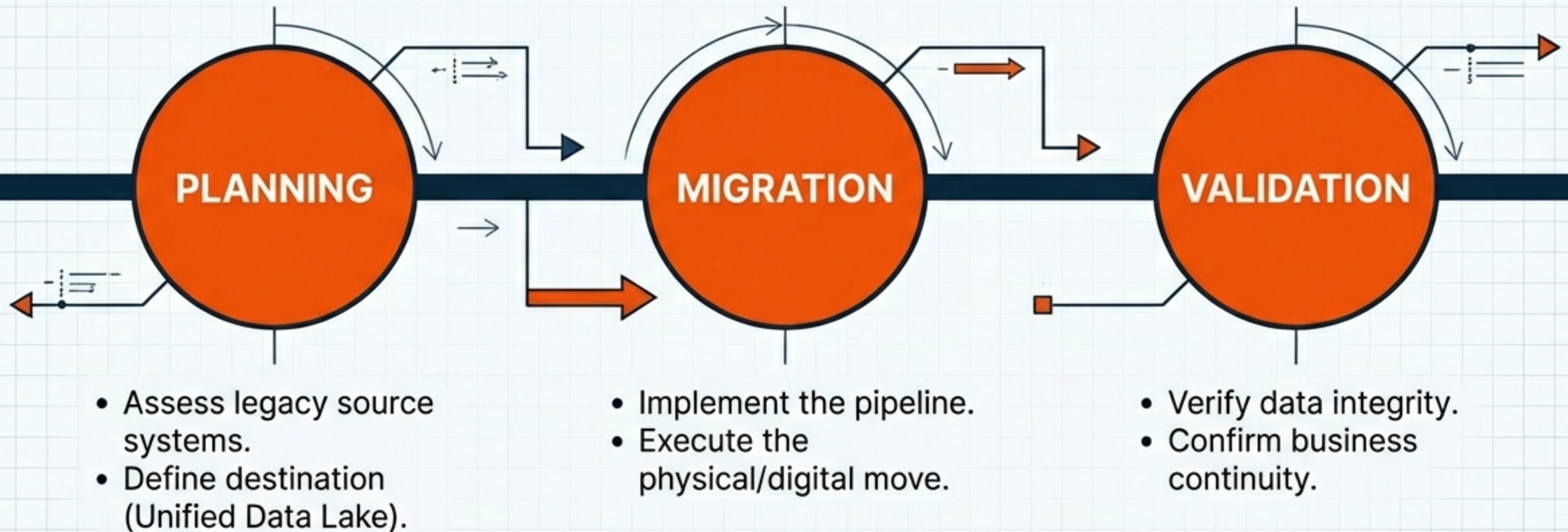


4. DATA CENTER MIGRATION



THE MIGRATION PROCESS: A PRACTICAL SCENARIO

Capstone Scenario: Junior Specialist for a Retail Chain.



TECHNICAL GLOSSARY & CHECKLIST

INTEGRATION	☒ ETL (Extract, Transform, Load)	☒ Connectors (APIs, Database Links)	☒ DIOM (Operations Mgmt)
STORAGE	☒ Block Storage (Performance)	☒ Object Storage (Scalability)	☒ RAID (Redundancy)
MIGRATION	☒ Legacy Systems	☒ Data Lake	☒ Validation Pipeline

BUILDING THE PROFESSIONAL SKILLSET

Content derived from: Data Integration, Data Storage, & Data Migration Course by SkillUp.

Part of the IBM Data Management Professional Certificate.

SKILLS ACQUIRED:

- Data Architecture & Pipelines
- Cloud Storage (Azure Simulation)
- Disaster Recovery & Security
- Legacy System Modernization



“In a digital economy, the ability to integrate, store, and migrate data is the foundation of business agility.”

SKILLUP | IBM DATA MANAGEMENT PROFESSIONAL CERTIFICATE

